

lab5-YUE CHENGHAO-227154

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code

- **RMIRegistryServer**

```
package com.example.lab5;

import java.rmi.RemoteException;
import java.rmi.registry.LocateRegistry;

public class RMIRegistryServer {
    public static void main(String[] args) throws RemoteException {
        LocateRegistry.createRegistry(1099);
        System.out.println("RMI Registry started on port 1099.");
        // 保持进程不退出
        while (true) {

            try { Thread.sleep(1000); } catch (InterruptedException ignored) {}
        }
    }
}
```

- **RegisterCircleServer**

```
package com.example.lab5;

import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;

public class RegisterCircleServer {
    public static void main(String[] args) throws Exception {
        String host = (args.length > 0) ? args[0] : "localhost";
        int port = (args.length > 1) ? Integer.parseInt(args[1]) : 1099;

        CircleService service = new CircleServiceImpl();

        Registry registry = LocateRegistry.getRegistry(host, port);
```

```

        registry.rebind("CircleService", service);

        System.out.println("CircleService bound in registry at " + host +
        ":" + port);
        System.out.println("Server is ready.");
        // keep the process
        while (true) {
            try { Thread.sleep(1000); } catch (InterruptedException ignored)
        {}
        }
    }
}

```

- **circleserve (interface)**

```

package com.example.lab5;

import java.rmi.Remote;
import java.rmi.RemoteException;

public interface CircleService extends Remote {
    double area(double radius) throws RemoteException;
}

```

- **CircleClientFX**

```

package com.example.lab5;

import javafx.application.Application;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.TextField;
import javafx.scene.layout.GridPane;
import javafx.stage.Stage;

import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;

public class CircleClientFX extends Application {

    private CircleService service;

    @Override

```

```

public void start(Stage stage) {
    // ===== UI =====
    Label title = new Label("RMI Circle Area Calculator");
    Label radiusLabel = new Label("Radius:");
    TextField radiusField = new TextField();
    radiusField.setPromptText("e.g. 2.5");

    Button calcBtn = new Button("Calculate Area");
    Label resultLabel = new Label("Area: -");
    Label statusLabel = new Label("Status: Not connected");

    GridPane root = new GridPane();
    root.setPadding(new Insets(16));
    root.setHgap(10);
    root.setVgap(10);

    root.add(title, 0, 0, 2, 1);
    root.add(radiusLabel, 0, 1);
    root.add(radiusField, 1, 1);
    root.add(calcBtn, 1, 2);
    root.add(resultLabel, 0, 3, 2, 1);
    root.add(statusLabel, 0, 4, 2, 1);

    // ===== RMI connect =====
    // 课上一般默认 localhost:1099; 你也可以改成读取 args
    String host = "localhost";
    int port = 1099;

    try {
        Registry registry = LocateRegistry.getRegistry(host, port);
        service = (CircleService) registry.lookup("CircleService");
        statusLabel.setText("Status: Connected to " + host + ":" +
port);
    } catch (Exception e) {
        statusLabel.setText("Status: Connect failed (" +
e.getClass().getSimpleName() + ")");
        // 不直接退出, 让你看到窗口和错误状态
    }

    // ===== Button action =====
    calcBtn.setOnAction(evt -> {
        if (service == null) {
            statusLabel.setText("Status: Not connected. Start Registry +
Server first.");
            return;
        }

        String input = radiusField.getText().trim();

```

```

        if (input.isEmpty()) {
            statusLabel.setText("Status: Please enter a radius.");
            return;
        }

        try {
            double r = Double.parseDouble(input);
            double area = service.area(r);
            resultLabel.setText("Area: " + area);
            statusLabel.setText("Status: OK");
        } catch (NumberFormatException nfe) {
            statusLabel.setText("Status: Invalid number.");
        } catch (Exception ex) {
            statusLabel.setText("Status: RMI error (" +
ex.getClass().getSimpleName() + ")");
        }
    });

    // ===== Stage =====
    stage.setTitle("RMI Lab - JavaFX Client");
    stage.setScene(new Scene(root, 420, 220));
    stage.show();
}

public static void main(String[] args) {
    launch(args);
}
}

```

- **output**

```
/Users/y/Library/Java/JavaVirtualMachines/liberica-full-25.0.1/bin/java ...  
CircleService bound in registry at localhost:1099  
Server is ready.
```

